	Target engagement	Dose	Efficacy	Safety
Metabolic	**mucopolysaccharidosis type 1** IDUA replacement / gene therapy (4)	**2020** NSG mice engrafted with human (incl. MPS I patient) HSPC… △ No lasting IDUA in patients	**2018** Rhesus monkey PK and chronic toxicology, AGT-181 Dosed to 6 mg/kg within margin	**2022** MPS I cat, intrathecal/intracisternal… Not measured in patients	**2020** NSG mice engrafted with human (incl. MPS I patient) HSPCs transduced with clinical-grade IDU…
	mucopolysaccharidosis type 3B NAGLU replacement / gene therapy (3)	**2021** Naglu-/- mouse, ICV NAGLU-IGF2 (tralesinidase alfa) △ CSF HS change small, not…	**2010** MPS IIIB dog, intracerebral rAAV5-hNAGLU with… No checkable prediction	**2021** Naglu-/- mouse, intraputaminal AAV2/AAV5-NaGlu Not measured in patients	**2022** MPS IIIB dog, ICV tralesinidase alfa (20 months) Patients had drug-related CNS SAEs
Neurology	**Duchenne muscular dystrophy** DMD (8)	**2011** DMD patient-derived myoblasts/myotubes and human muscle cell lines (sequence…	**2006** mdx mouse dosed with mouse-specific exon 23 PMO… No checkable prediction	**2013** DMD patient-derived myoblasts/myotubes and human… ☐ Outcome trial negative	**2019** Rodent and cynomolgus… Predicted kidney toxicity absent in boys
	Huntington disease HTT antisense / siRNA (4)	**2022** HD patient iPSC-derived neurons and astrocytes (HD71, HD180) on AAV5-miHTT	**2019** HD rodents and tgHD minipig (AAV5-miHTT dose response) No checkable prediction	**2025** HD rodents (Q175 knock-in, R6/2) on intrastriatal AAV5-miHTT △ cUHDRS worse on tominersen	**2022** NHP and rat intrastriatal AAV5-miHTT toxicology High-dose inflammatory SUSARs
	fragile X syndrome GRM5 negative modulators (3)	**2009** Fmr1 KO mouse prepulse inhibition (PPI) of startle	**2005** Fmr1 KO mouse behaviour and spine rescue No checkable prediction	**2014** Drosophila dfmr1 mutant… Basimglurant missed ADAMS primary	**2008** Sponsor GLP toxicology (rodent and non-rodent) Model data unpublished
	primary erythermalgia SCN9A blockers (3)	**2016** Patient iPSC-derived sensory neurons (IEM) Not measured in patients	**2016** PBPK model from a human IV/oral microdose study (PF-05089771)	**2016** SCN9A human genetics (loss-of-function CIP, gain-of-function IEM)	**2011** Non-clinical toxicology (PF-05089771, funapide) Model data unpublished
Oncology	**B-cell non-Hodgkin lymphoma** MS4A1-directed depleters (3)	**2024** MB-CART20.1 product co-culture with CD20+ target cells No checkable prediction	**2005** Disseminated Daudi lymphoma in SCID mice No checkable prediction	**2024** MB-CART20.1 product co-culture with CD20+ target cells No checkable prediction	**2005** Disseminated Daudi lymphoma in SCID mice No checkable prediction
	Waldenstrom macroglobulinemia MS4A1-directed depleters (3)	**2009** Cynomolgus monkeys dosed with ofatumumab (B-cell depletion… Human B cells fell as in monkeys	**2012** SCID mice bearing CD20+ human B-cell tumour xenografts dosed with ofatumumab (tumour…	**2014** WM patient NK cells, ex-vivo ADCC against autologous WM B… Not measured in patients	**2010** Cynomolgus monkeys dose… WM harms matched monkeys; retrodiction
	acute lymphoblastic leukemia CD19-directed depleters (6)	**2013** 19-28z CAR T cells in SCID-Beige mice with systemic NALM-6 ALL	**2013** Patient-derived ALL xenografts in NOD/SCID mice… No checkable prediction	**2021** Syngeneic murine anti-mouse-CD19 CAR T cells in… △ ORR 25.5%, trial stopped	**2021** Syngeneic murine anti-mouse-CD19 CAR T cells in lymphodepleted lymphoma-bearin…
	chronic lymphocytic… BTK inhibitors (4)	**2025** QSP model of tissue BTK… QSP reproduced acalabrutinib occupancy	**2022** Sponsor PBPK model (Simcyp) for CYP3A drug-interaction dosing	**2019** Spontaneous canine B-cell lymphoma (pet dogs) ASCEND PFS positive	**2021** Kinase selectivity panel and ex-vivo platelet/NK assays vs… △ Tox findings not compared in…
	chronic lymphocytic… CD19-directed depleters (4)	**2014** Patient CLL cells with NK effectors (ex-vivo ADCC) △ Ibrutinib did not raise CAR-T…	**2020** Cynomolgus monkey repeat-dose toxicology (tafasitamab) Margin held: no MTD up to 12 mg/kg	**2013** CD19 CAR-T in immunodeficient-mouse CD19+… △ No ORR gain over rituximab	**2020** CAR-T plus concurrent ibrutinib in CLL/ALL xenografts
	chronic lymphocytic… MS4A1-directed depleters (4)	**2023** Patient CLL PBMC co-culture and CLL patient-derived xenograft Not measured in patients	**2020** Cynomolgus monkey (B-cell depletion, cytokines, SC vs IV) No checkable prediction	**2023** Patient CLL PBMC co-culture and CLL patient-derived xenograft	**2021** Cynomolgus monkey (B-cell depletion, cytokines, SC vs IV)
	chronic myeloid leukemia ABL1 inhibitors (3)	**2012** BCR-ABL cell lines and mouse leukemia/xenograft models	**2012** Ba/F3 BCR-ABL cell-based mutagenesis screen	**2014** Ba/F3 BCR-ABL cell-based mutagenesis screen	**2025** Biochemical kinase-selectivity profiling (VEGFR2 and other… △ Heart, liver, pancreas flagged;…
	diffuse large B-cell lymphoma CD19-directed depleters (4)	**2017** Primary pre-B ALL xenograft in NSG mice (CTL019 CAR)	**2019** Cynomolgus monkey repeat-dose toxicology… No checkable prediction	**2017** Primary pre-B ALL xenograft in NSG mice (CTL019 CAR) △ No better than rituximab	**2014** Human lymphoma xenografts incl. DLBCL in SCID mice (SAR3419)
	diffuse large B-cell lymphoma MS4A1-directed depleters (5)	**2020** Cynomolgus monkey… Depletion known from anti-CD20 patients	**2020** Mechanistic QSP model of CD20xCD3 CRS (mosunetuzumab)	**2026** Humanized-mouse lympho… Phase 3 missed OS; surrogate already known	**2020** Cynomolgus monkey (epcoritamab toxicology/PD)
	extranodal nasal NK/T cell… anti-PDCD1 antibodies (3)	☐ no model relied on	☐ no model relied on	**2020** ENKTL tumour biopsies: EBV-driven PD-L1 expression (IHC)	**2015** Cynomolgus monkey repeat-dose toxicology (tislelizumab) No checkable prediction
	gastrointestinal stromal tumor HSP90AA1/HSP90AB1 inhibitors (3)	**2014** GIST-PSW and GIST430 xenografts in SCID mice (onalespi… Patient Hsp70 induced in tumour	**2012** Human tumour xenograft mice (non-GIST) dosed with BIIB021: tumour and serum Hsp70,…	**2012** Imatinib-sensitive and -resistant GIST cell lines (GIST882… △ Limited activity in TKI-resistant…	**2016** GIST-PSW and GIST430 xenografts in SCID mice (onalespib ± imatinib)
	large B-cell lymphoma CD19-directed depleters (3)	**2010** Syngeneic mouse lymphoma with anti-murine-CD19 CAR T cells	**2010** Syngeneic mouse lymphoma with anti-murine-CD19 CAR T cells No checkable prediction	**2016** Syngeneic mouse lymphoma with anti-murine-CD19 CAR T cells	**2010** Syngeneic mouse lymphoma with anti-murine-CD19 CAR T cells
	mantle cell lymphoma BTK inhibitors (4)	**2009** BTK active-site probe occupancy in mice, rats and dogs	**2012** BTK active-site probe occupancy in mice, rats and dogs	**2011** Pet dogs with spontaneous B-cell non-Hodgkin lymphoma	**2019** Zanubrutinib GLP toxicolog… Bleeding flagged, thrombocytopenia missed
	mantle cell lymphoma CD19-directed depleters (3)	**2020** Syngeneic mouse lymphoma, murine anti-mCD19 CAR T cells	**2021** Rat and cynomolgus monkey toxicology (loncastuximab tesirine) Model data unpublished	**2019** Syngeneic mouse lymphoma, murine anti-mCD19 CAR T cells	**2021** Rat and cynomolgus monkey toxicology (loncastuximab tesirine)
	marginal zone lymphoma MS4A1-directed depleters (3)	**2008** Ex-vivo ADCC on patient leukaemia/lymphoma cells (ublituximab vs rituximab)	**2026** Semi-mechanistic PBPK/PD model of epcoritamab trimer formation and tumour response…	**2018** Humanized-mouse lymphoma xenografts (CDX/PDX) Patient result unclear	**2022** IgM CD20xCD3 (imvotamab/IGM-2323 and cyno surrogate) in cynomolgus monkey…
	myelodysplastic syndrome HDAC (pan) inhibitors (3)	**2017** Mouse xenografts (HCT-116, later MOLM-13/SET-2 AML) Acetylation did not track exposure	**2011** Allometry and Simcyp PBPK (pracinostat) No checkable prediction	**2015** Mouse xenografts (HCT-116, later MOLM-13/SET-2 AML) CR 18% vs 33% placebo	**2010** Mouse xenografts (HCT-116, later MOLM-13/SET-2 AML) No checkable prediction
	myelofibrosis JAK2 inhibitors (3)	**2019** Patient-derived JAK2V617F+ progenitor colonies and xenograft (TG101348)	**2008** JAK2V617F retroviral bone-marrow transplant mouse No checkable prediction	**2013** Patient-derived JAK2V617F+ progenitor colonies and xenograft… △ Patients responded; mouse said…	**2008** JAK2V617F retroviral bone-marrow transplant mouse △ Assay missed thiamine harm po…
	small cell lung carcinoma anti-PDCD1 antibodies (3)	**2023** In-vitro PD-1 binding,… Patient troughs 8-12x the saturating level	**2015** Cynomolgus tox, nivolumab	**2021** Tumour-mutational-burden rule, SCLC exomes, nivolumab △ Missed OS, indication withdrawn	**2015** PD-1 knockout mice △ Fatal pneumonitis, not in monkeys
Ophthalmology	**diabetic macular edema** anti-VEGFA antibodies (3)	**2009** STZ-diabetic rat, Evans-blue blood-retinal-barrier leak	**2011** First-order decay model of intravitreal VEGF-binding activity built on rabbit half-lives	**2011** STZ-diabetic rat, Evans-blue blood-retinal-barrier leak	**2023** Repeat-dose intravitreal… PHOTON: no nasomucosal signal at 8 mg
	dry eye syndrome anti-TNF antibodies (3)	**2023** Aged mouse dry eye Not measured in patients	**2013** Rabbit topical ocular PK (ESBA105 eye drops)	**2020** Canine dry eye (spontaneous) Dog sign benefit; Phase 3 flat vs placebo	**2022** Rabbit topical ocular PK (ESBA105 eye drops)
	retinitis pigmentosa RPGR replacement / gene therapy (3)	**2024** Isogenic RPGR-knockout human iPSC retinal organoids,… No checkable prediction	**2024** XLPRA2 RPGR-mutant dogs, AGTC-501 (laru-zova)	**2026** XLPRA2 RPGR-mutant dogs, AGTC-501 (laru-zova) △ LUMEOS missed VMA primary	**2024** XLPRA2 RPGR-mutant dogs, AGTC-501 (laru-zova) △ Inflammation at higher doses,…
Psy.	**insomnia** HCRTR1/2 antagonists (6)	**2013** Rat, dog and rhesus sleep… Healthy men slept more, as animals did	**2022** 13-week dog toxicology: cataplexy NOAEL (daridorexant)	**2012** Rat, dog and rhesus sleep… Insomnia patients slept more, as predicted	**2019** Mouse emotion-induced cataplexy (lemborexant)
Respiratory	**cystic fibrosis** CFTR correctors (10)	**2024** FRT cells expressing CFTR variants △ Third corrector added nothing	**2015** Rat and dog toxicology (lumacaftor)	**2024** FRT cells expressing CFTR variants	**2015** Rat and dog toxicology (lumacaftor)
	cystic fibrosis CFTR potentiators (5)	**2026** Patient-derived primary nasal epithelial cells (HNE, Ussing chamber theratyping)	**2022** Human hepatic metabolism in vitro, scaled to PK	**2025** Patient-derived intestinal organoids (forskolin swelling) Patient result unclear	**2024** Rat toxicology (juvenile and adult, ivacaftor program)
	pulmonary arterial hypertension PTGIR agonists (3)	**2019** Rat MCT and Su5416/hypoxia, intratracheal… Patient result unclear	**2016** Rat MCT and Su5416/hypoxia, intratracheal… Model data unpublished	**2019** PAH-patient PASMC proliferation and human distal pulmonary artery myography ex vi…	**2002** Monocrotaline rat (oral dosing)
Uro.	**interstitial cystitis** anti-NGF antibodies (3)	**1997** Rodent chemical cystitis Not measured in patients	**1997** Rodent chemical cystitis No checkable prediction	**2011** Rodent chemical cystitis △ ASP6294 missed IC pain endpoint	**2012** Primate and rodent joint tox △ Predicts joint harm IC patients…
Other	**atopic eczema** HRH4 antagonists (3)	**2014** Mouse FITC dermatitis, H4R	**2010** Mouse FITC dermatitis, H4R No checkable prediction	**2021** Dog acute atopic dermatitis, JNJ 7777120 and JNJ 28307474 △ Adriforant: mice yes, patients no	**2014** Rat, monkey 6/9-month tox Agranulocytosis the tox species never showed